Jackie (Jingkai) Shao

Waterloo, Ontario | (+1)514-562-8684 | j66shao@uwaterloo.ca | linkedin.com/jackie | portfolio.com/jackie | website.com/jackie

Work Experience

Manufacturing Controls Engineering Intern **April 2026 – May 2026**
OPmobility (ex. Plastic Omnium) *Exchange Program*

- Developed and deployed KRL control systems on the KRC4 controller for filament winding applications on Type IV hydrogen mobility vessels, improving winding accuracy around the dome of the thermoplastic liner by 15%.
- Generated aesthetically simple HMI screens in the TIA Portal, reducing operator error rate by 10% while providing sufficient feedback for troubleshooting through structured alarm hierarchies.
- Developed an automation program using Python (pandas + NumPy) that parses ComposiCAD output files and cross-references with actual winding parameters from PLC Memory to flag deviations, cutting inspection time by 30%.

Mechanical Design Engineering Intern **December 2025 – March 2026**
Baker Hughes (ex. Chart Industries, Inc.) *Co-op Program*

- Designed 10+ SolidWorks 3-D models using DFM/DFA principles and AutoCAD 2-D drawings (Part and Assembly) for the bulk storage tanks, improving the First Pass Yield (FPY) from 78% to 85%.
- Designed 5+ inner pipeline 2-D layouts in AutoCAD, improving model clarity for the sloshing-induced loads calculation and cutting simulation setup time by 35%.
- Developed MIG/TIG welding procedures for external tank structures, redesigning joint geometries to improve weld accessibility and reduce weld time per assembly by 10%.

System Design & Automation Engineering Intern **May 2025 – August 2025**
OPmobility (ex. Plastic Omnium) *Exchange Program*

- Designed 5+ CATIA 3D models using DFM/DFA and 2D drawings (Part and Assembly) with GD&T for the thermoplastic liners, resulting in better compatibility with automated manufacturing.
- Conducted PLC programming (ladder logic) on the Siemens S7-1500 to reduce the wedge effect of the composite filament layers by developing a Tension Profile, improving manufacturing consistency from $C_{pk} < C_p$ to $Cpk \geq 1.33$.
- Performed non-linear FEA in Abaqus using raw data from proprietary ComposiCAD software to pinpoint structural weaknesses, predict burst pressure, and simulate pressure cycle testing, reducing prototyping iteration time by 30%.
- Interfaced Siemens S7-1500 PLC with conveyors, KUKA robots, and Manufacturing Execution System (MES) over PROFINET, reducing inter-device communication faults by 10%.
- Implemented control reliable safety systems for safeguarding compliant with ANSI/RIA 15.06 and OSHA standards, integrating dual-channel E-stops, light curtains, and safety relays across 5+ machine cells.

Projects

Filament Winding Robotic Arm Digital Twin (KUKA.Sim) | *Github* **Associated with OPmobility**

- Developed a Digital Twin Robotic Arm in KUKA.Sim to perform a Gage R&R study on fibre winding angle, improving the measurement accuracy of the rotary encoders by 10%.
- Reduced physical rework by 10+ hours by validating robot programs in KUKA.Sim digital twin prior to deployment, catching 5+ logic errors during pre-commissioning.

Tooling Design (Jigs & Fixtures) | *Github* **Associated with UWaterloo**

- Designed a jig with spring-loaded hold-down clamps (DFA) in SolidWorks, eliminating the need for threaded fasteners, which reduces the part assembly time by 20%.
- Designed a fixture with single-motion clamping instead of a screw-down clamp (DFM), eliminating the need for precision-threaded holes in the fixture body, which reduces manufacturing cost by 10%.

Education

Bachelor of Applied Science in Mechanical Engineering **September 2025 - June 2030**
University of Waterloo, Honours Program with Co-op | GPA: 3.7/4.0 *Ontario, Canada*

- **Relevant Coursework:** Engineering Design, C++ Programming, Material Science, Electrical Circuits, Mechanics
- **Student Activities:** System Design Engineering Researcher at the Waterloo Institute for Sustainable Aeronautics

Skills

Software: CATIA, SolidWorks, Abaqus, Ansys, AutoCAD, Triflex, Robotstudio, KUKA.Sim, Word, Excel, PowerPoint, Visio
Technical: Analog and Digital Circuits, Control Systems, Logic, Industrial Robots (Kuka, ABB), Servo Controllers (SINAMICS S120)
Safety: NFPA79/NEC/UL508A, ANSI/RIA15.06/OSHA, ASME BPVC Section XII, ASME Section VIII, ASME B31.3 (Process Piping)
Manufacturing: CNC Machining, Common Power and Hand Tools, Manual Lathe/Mill, Filament Winding, Blow/Injection Moulding, Welding
Programming: RAPID, KRL, C++, Python, MATLAB/Simulink, Ladder Logic, Git